- Sporangia, spore bearing bag like structures are borne on special leaves called sporophyll. The Sporophylls get organized to form cone or strobilus. Example: *Selaginella*, *Equisetum* .
- They may be **homosporous** (produce one type of spores-*Lycopodium*) or **Heterosporous** (produce two types of spores-*Selaginella)*. Heterospory is the origin for seed habit.
- Development of sporangia may be **eusporangiate** (development of sporangium from group of initials) or **leptosporangiate** (development of sporangium from single initial).
- Spore mother cells undergo meiosis and produce spores (n).
- Spore germinates to produce haploid, multicellular green, cordate shaped independent gametophytes called prothallus.
- Fragmentation, resting buds, root tubers and adventitious buds help in vegetative reproduction.
- Sexual reproduction is oogamous. Sex organs, namely antheridium and archegonium are produced on the prothallus.
- Antheridium produces spirally coiled and multiflagellate antherozoids.
- Archegonium is flask shaped with broad venter and elongated narrow neck. The venter possesses egg or ovum and neck contain neck canal cells.
- Water is essential for fertilization. After fertilization a diploid zygote is formed and undergoes mitotic division to form embryo.
- Pteridophytes show **apogamy** and **apospory.**

2.5.2 Classification of Pteridophytes

Reimer (1954) proposed a classification for pteridophytes. In this classification, the pteridophytes are divided into five subdivisions. 1. Psilophytopsida 2. Psilotopsida 3. Lycopsida 4. Sphenopsida 5. Pteropsida. There are 19 orders and 48 families in the classification.

2.5.3 Economic Importance

The Economic importance of Pteridophyte is given in Table 2.4

Table 2.4: Economic importance of Pteridophyte

Pteridophyte	Uses
Rumohra adiantiformis (leather leaf fern)	Cut flower arrangements.
Marsilea	Food
Azolla	Biofertilizer.
Dryopteris filix–mas	Treatment for tapeworm.
Pteris vittata	Removal of heavy metals from soils - Bioremediation
Pteridium sp.	Leaves yield green dye.
Equisetum sp.	Stems for scouring.
Psilotum, Lycopodium, Selaginella, Angiopteris, Marattia	Ornamental plants

2.5.4 Selaginella

Division – Lycophyta
Class – Ligulopsida
Order – Selaginellales
Family –Selaginellaceae
Genus – *Selaginella*

Selaginella is commonly called 'spike moss'. They are distributed in humid temperate and tropical rain forests. *Selaginella rupestris* and *Selaginella lepidophylla* are Xerophytic. *Selaginella kraussiana*, *Selaginella chrysocaulos, Selaginella megaphylla* are some common species. In few *Selaginella* species during dry season the entire plant body gets curled and become fresh, green when moisture is available. Due to this they are called **Resurrection plants**. Example *S. lepidophylla*

External morphology

Habit

The plant body of *Selaginella* is sporophyte (2n) and it is differentiated into root, stem, and leaves (Figure 2.16). There exist variations in the habit of *Selaginella*. Some species possess

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